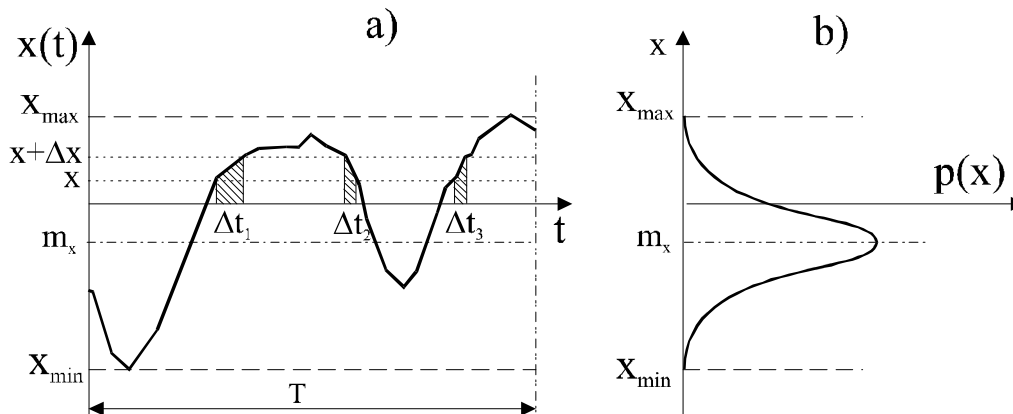


## Exercise 2

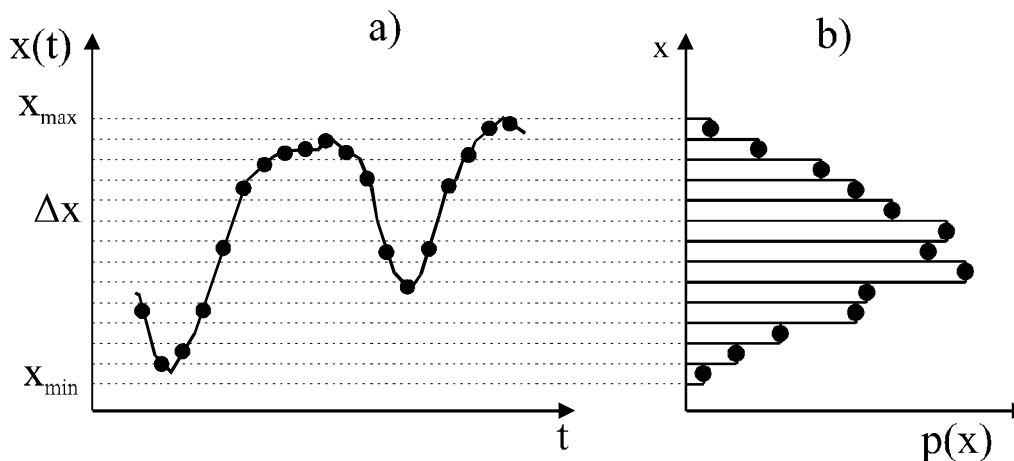
### Probability density function (PDF)

$$p(x_0) = \lim_{\Delta x \rightarrow 0} \frac{P(x_0 < x \leq x_0 + \Delta x)}{\Delta x}$$



$$P(x \in \langle x_0; x_0 + \Delta x \rangle) = \frac{\sum \Delta t_i}{T}; \quad p(x_0) = \frac{P}{\Delta x}$$

numerical algorithm for pdf estimator



$$p(x_0) = \frac{P}{\Delta x} = \frac{n/N}{\Delta x}$$

$n$  - number of "supporting" samples (seen through the window " $\Delta x$ ")

$N$  - total number of samples

### **Compulsory tasks**

- a) perform stationarity test for either random or compound signal (determine the signal length ensuring sufficient accuracy of PDF distribution); if the test is conducted for compound signal the noise content should be at least 50%
- b) determine the optimal number of intervals (bins) ensuring smooth PDF distribution for the same signal as analysed in task **a**
- c) determine the PDF distributions for typical deterministic signals
- d) analyse the evolution of a PDF distribution for selected compound signal changing the level of noise content (see appendix) in the range 0-100%

### **Additional (optional) tasks**

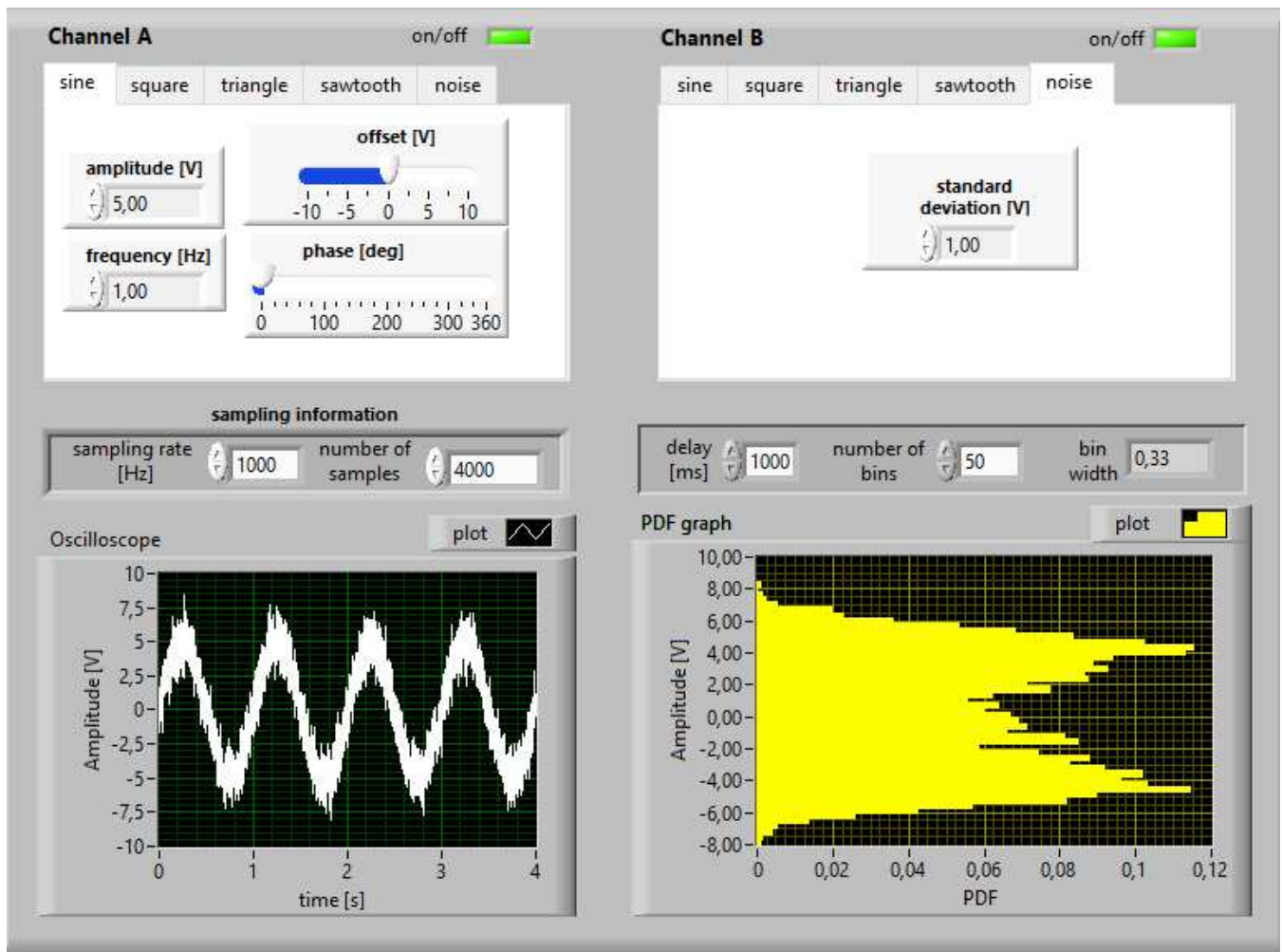
- determine the influence of the signal's phase angle, amplitude, offset and frequency on the PDF distribution for a periodic signal
- .....
- .....



### **The exercise aims at learning how to:**

- **conduct correctly pdf analysis (selection of signal duration and number of pdf bins),**
- **extract the information about the signal (type, composition and parameters) from pdf distribution (of course, not seeing the signal).**

## project "PDF" in LabView environment



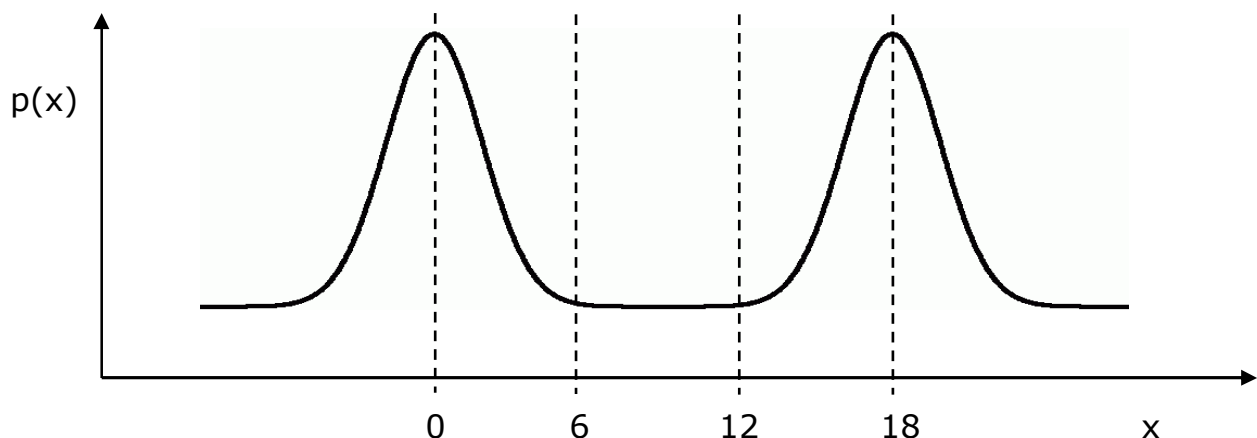
## Sample problems

1. Draw the pdf distributions for the following signals:

- random (Gaussian) noise,
- sine wave,
- square wave,
- sawtooth wave,
- disturbed (noisy) sine wave.

Please indicate the relations between signal parameters and corresponding pdf distributions.

- Discuss the possible sources of the distortions (uncertainties) of the pdf distributions. Give examples.
- What is the influence of the frequency and the phase angle of a square wave on its pdf distribution ?
- Determine the composition of the signal having the following pdf distribution (see the graph). Try to estimate the signal parameters.



- Draw the pdf distribution of the selected compound signal and explain/discuss its shape.
- What are the differences between pdf distributions corresponding to the sine waves shown below ? Compare them graphically.
  - $x_1 = 3 + A \cdot \sin(2\pi f \cdot t + \varphi)$   
 $x_2 = 5 + A \cdot \sin(2\pi f \cdot t + \varphi)$
  - $x_1 = x_0 + 2 \cdot A \cdot \sin(2\pi f \cdot t + \varphi)$   
 $x_2 = x_0 + 4 \cdot A \cdot \sin(2\pi f \cdot t + \varphi)$
  - $x_1 = x_0 + A \cdot \sin(2\pi f \cdot t + \varphi)$   
 $x_2 = x_0 + A \cdot \sin(2\pi (2 \cdot f) \cdot t + \varphi)$
  - $x_1 = x_0 + A \cdot \sin(2\pi f \cdot t + \varphi)$   
 $x_2 = x_0 + A \cdot \sin(2\pi f \cdot t + (\varphi + \pi))$

## Appendix

Let's consider the signal  $X(t)$ . Its power is given as

$$N = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T X^2(t) dt$$

When the signal composition is known, then the power may be expressed as a sum of partial powers  $N_i$

$$N = \sum_i N_i$$

For instance, if the signal is composed of a sine wave with a non-zero mean value  $\bar{X}$  and disturbed by a noise then

$$N = N_{\sin} + N_{\text{noise}} + N_{\text{mean}} = \sigma_{\sin}^2 + \sigma_{\text{noise}}^2 + \bar{X}^2 = \frac{A^2}{2} + \sigma_{\text{noise}}^2 + \bar{X}^2$$

where

$\sigma_{\sin}$  - standard deviation of the sine wave

$A$  - amplitude of the sine wave

$\sigma_{\text{noise}}$  - standard deviation of the noise

The content of the noise in the signal can be calculated as follows

$$\varepsilon = \frac{N_{\text{noise}}}{N - N_{\text{mean}}} \cdot 100\%$$